21 Show that $\dfrac{6-\sqrt{8}}{\sqrt{2}-1}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

$\sqrt{8} = 2\sqrt{2}$. Multiply by the conjugate:

$$\frac{6 - 2\sqrt{2}}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$$

1 mark
Multiply top and bottom by $\sqrt{2} + 1$

×	$\sqrt{2}$	$+1$
6	$6\sqrt{2}$	$+6$
$-2\sqrt{2}$	-4	$-2\sqrt{2}$

1 mark
Expand the numerator completely

Denominator: $2 - 1 = 1$

Numerator: $6\sqrt{2} + 6 - 4 - 2\sqrt{2} = 2 + 4\sqrt{2}$.

Dividing by 1 gives $2 + 4\sqrt{2}$, so $a = 2$ and $b = 4$.

1 mark
Correct integer coefficients

(Total for Question 21 is 3 marks)